Utkarsh Sharma

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SUMMARY

Data engineer and technologist pursuing a Master of Science in Applied Data Intelligence at SJSU, with a Bachelor's in Computer Science. Skilled in building scalable ETL pipelines, cloud-native infrastructure, and data workflows using Python, SQL, AWS, dbt, and Apache tools. Familiar with AI agents, large language models, and intelligent systems — driven to apply emerging AI technologies to solve real-world data problems at scale.

EDUCATION

San Jose State University — Jan 2025 – Dec 2027 (Expected)
Master of Science in Applied Data Intelligence

Relevant Coursework: Data Warehouse, Distributed Systems, Data Visualization, Database Tech for DA, Math Methods for DI, Programming for DA

California State University, East Bay — 2020 – 2023
Bachelor of Science in Computer Science

SKILLS & TOOLS

Programming & Languages: Python, SQL, R
Data Engineering & ETL: Apache NiFi, dbt, AWS S3, Amazon Redshift, Google Cloud Platform, Apache Kafka, Docker
Databases: MySQL, MongoDB, Neo4j, PostgreSQL
Data Visualization: Tableau, Power BI, Matplotlib, Seaborn
Machine Learning & AI: Scikit-learn, Pandas, NumPy, LLMs, AI Agents
Tools & Platforms: Git/GitHub, Jupyter Notebook, Linux, Kubernetes, Advanced Excel (Pivot Tables, VBA, XLOOKUP)

WORK EXPERIENCE

Total Transportation Solutions — Jul 2023 – Sep 2024
IT Technician Assistant

- Analyzed operational performance data across 50+ users, identifying patterns and trends to reduce recurring incidents by 30% using data-driven root-cause analysis.
- Tracked and reported on system performance metrics across 15+ cases, developing dashboards and summaries to support management decision-making.
- Maintained data integrity and access controls for internal reporting systems, ensuring data quality for operational analytics.
- Generated automated performance reports combining data from multiple sources to improve operational efficiency.

PROJECTS

NBA Player Value Warehouse Pipeline -- Python, Apache NiFi, AWS S3, Amazon Redshift, Tableau

- Built an end-to-end ETL pipeline ingesting 5 seasons of NBA player stats and salary data from Basketball Reference and Kaggle, cleaning and loading into Amazon Redshift via Apache NiFi and Python.
- Designed a star schema data warehouse with a central fact table and 3 dimension tables (player, team, season) optimized for analytical queries.
- Engineered a linear regression model using 7 advanced metrics (PER, VORP, WS, BPM, TS%, USG%, MP) to classify players as overpaid, fair value, or underpaid.
- Delivered an interactive Tableau dashboard enabling salary vs. performance comparisons filterable by player, team, and season.

RevU — Movie Review Desktop App -- Python, PyQt5, MySQL, Matplotlib

- Developed a full-stack desktop application with role-based authentication routing users to a customer portal or admin dashboard via live MySQL queries.
- Implemented movie/director search, genre-filtered top-10 charts, and a random movie picker using SQL JOINs across a 4-table MySQL schema.
- Built an admin analytics dashboard with embedded Matplotlib charts for monthly user signups and gross revenue trends.

COMPAS Algorithmic Fairness Analysis -- Python, Pandas, Scikit-learn, Statistical Modeling

- Cleaned and merged 60,843 raw COMPAS assessments into an 11,189-row analytic dataset (98.6% merge success rate).
- Applied chi-square, Welch's t-test, and Mann–Whitney U tests — finding African-American defendants averaged 1.25 points higher on the COMPAS scale ($p < 0.001$).
- Built multiple regression and Random Forest models confirming race as a significant predictor ($\beta \approx 1.40$) even after controlling for age and prior convictions.